

TASK 3 – WORD COUNTER

Objective

The objective of this project is to create a Python program that reads a text file and counts the number of words present in it. The program also provides additional statistics such as line count, character count, and unique word count.

Description

This Word Counter application is developed using Python. The program reads the contents of a text file, processes the text, and calculates various statistics. It also handles exceptions such as missing files and invalid inputs.

Tools and Technologies Used

- Python Programming Language
- Google Colab / VS Code
- Text File (.txt)

Features

- Reads data from a text file
- Counts total number of words
- Counts total number of lines
- Counts total number of characters
- Calculates unique words
- Displays the most frequent words
- Handles file-related errors using exception handling

Program Workflow

1. Accept the file name from the user.
2. Open the file in read mode.
3. Read the contents of the file.
4. Split the text into words.
5. Count words, lines, and characters.
6. Display the results.
7. Handle exceptions if the file is not found.

Output

The program successfully analyzes the text file and displays:

- Total Words
- Total Lines

- Total Characters
- Unique Words
- Most Frequent Words

Conclusion

The Word Counter project demonstrates the use of file handling, string manipulation, and exception handling in Python. It is a simple yet effective application for analyzing text files and extracting useful information.